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Efficient leukocytes detection and classification in microscopic blood images using convolutional neural network coupled with a dual attention network

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ABSTRACT

Leukocytes, also called White Blood Cells (WBCs) or leucocytes, are the cells that play a pivotal role in human health and are vital indicators of diseases such as malaria, leukemia, AIDS, and other viral infections. WBCs detection and classification in blood smears offers insights to pathologists, aiding diagnosis across medical conditions. Traditional techniques, including manual counting, detection, classification, and visual inspection of microscopic images by medical professionals, pose challenges due to their labor-intensive nature. However, traditional methods are time consuming and sometimes susceptible to errors. Here, we propose a high-performance convolutional neural network (CNN) coupled with a dual-attention network that efficiently detects and classifies WBCs in microscopic thick smear images. The main aim of this study was to enhance clinical hematology systems and expedite medical diagnostic processes. In the proposed technique, we utilized a deep convolutional generative adversarial network (DCGAN) to overcome the limitations imposed by limited training data and employed a dual attention mechanism to improve accuracy, efficiency, and generalization. The proposed technique achieved overall accuracy rates of 99.83%, 99.35%, and 99.60% for the peripheral blood cell (PBC), leukocyte images for segmentation and classification (LISC), and Raabin-WBC benchmark datasets, respectively. Our proposed approach outperforms state-of-the-art methods in terms of accuracy, highlighting the effectiveness of the strategies employed and their potential to enhance diagnostic capabilities and advance real-world healthcare practices and diagnostic systems.

1. Introduction

Medical image analysis (MIA) involves utilizing a range of computer software and algorithms to process, interpret, and extract significant information from medical images [1,2]. Hematologic diagnostic systems in the field of medical imaging are based on the study of white blood cells (WBCs) in blood microscopic images [3]. The human immune system is primarily composed of WBCs, which play a crucial role in the fight against various bacterial and viral diseases [4]. Blood plays a vital and indispensable role in the human body [5], and the normal percentage of blood in the human body is approximately 7% of the body weight of younger individuals [6]. Blood consists of three component, platelets (thrombocytes), WBCs and red-blood cells (RBCs). Platelets are tiny cell fragments that contribute to the blood-clotting process [7]. RBCs have the crucial function of carrying oxygen from the lungs to every part of the body, while WBCs defend the body against infections and foreign substances. WBCs contain granules that are further categorized based on their physical structure and function into lymphocytes, monocytes, basophils, eosinophils, and neutrophils [8]. Additionally, an important feature of WBCs is counting and variation in the percentage between subtypes [9] which helps in determining the severity of infection, blood illness, and diseases such as anemia [10], leukemia [11], and

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malaria [12]. Therefore, WBC analysis plays a significant role in biomedical procedures and assists medical experts in diagnosing different hematological problems. In most cases, it is important to classify different types of WBCs in thick blood smear images to diagnose and treat several health conditions [13]. Moreover, each WBC subclass has normal morphological properties (shape, size, color, and texture) and a differential count. Based on these properties, medical experts can differentiate between cells from one another. WBC subclasses have distinct properties; however, some similarities exist among them in terms of size, shape, texture, and color. Detailed information and sample images of each WBCs subclass are provided in Table 1. WBC counts provide important information to doctors for diagnosing various hematic diseases and other health conditions [14]. Structural variations and fluctuations in the differential count % of WBCs in blood represent various health disorders [15]. WBC analysis provides a large amount of diagnostic information to medical experts for the analysis of WBCs in smear images. The count, size, and morphological shape of WBCs provide valuable information to health experts for the diagnosis of numerous hematic diseases [16].

The deformability of the WBC subtype lymphocyte cell shape is a perfect indication of acute lymphoblastic leukemia (ALL), and the shape of each WBC subtype presents many diseases, including HIV, AID, blood cancer, and other viral and bacterial diseases [17]. Therefore, WBCs play a vital role in accurate early disease diagnosis and assist medical experts in the correct treatment of patients [1,16]. The WBC analysis of smear images assists medical experts in the diagnosis of various diseases, including blood cancer, AIDS, and other hematic disorders, by analyzing the size, count, and shape of WBCs in smear images [18]. The detection and identification of WBCs in their subcategories are complicated because of their color, size, and shape diversity. Typically, manual counting is used for such tasks; however, it is a laborious, challenging, and time-consuming task for medical experts and is sometimes prone to errors. With the advancement of technology, several computer-aided techniques were used for counting WBCs, such as automated hematology analyzers, image analysis, flow cytometry, morphology recognition, digital microscopy, and artificial intelligence (AI) algorithms [15]. These techniques have significantly improved the performance of WBCs analysis on blood smear images. However, detection and classification are still challenging [19]. These issues include low image quality and a lack of freely accessible large and high-quality medical datasets. Image quality influences the appearance of cells and can be influenced by features such as staining, focus and lighting. Cell overlapping is also an issue because in some instances, multiple WBCs may appear in one microscopic image making it hard to correctly segment and classify sub-class regions. Class imbalance happens when different types of WBCs in a blood sample have similar inner subclass shape and structure similarities [20]. Different human annotators may use different criteria for identifying WBC leading to inconsistent labels used to train the model. In addition, there may be a limited amount of annotated data available for training and evaluating a model, which would impact overall performance. The challenges faced pose difficulties for computer-aided systems to accurate and efficient identification and categorization of WBCs in microscopic images. It is thus possible for researchers to use various methods that can help improve the quality of images. This may involve employing class-balancing techniques to address imbalances in classes and utilizing image-preprocessing methods to enhance image quality while expanding the size of the annotated dataset.

Table 1
WBC subtypes, their normal differential count, and morphological properties.

WBC sub-type		Count (%)	Morphological characteristics of WBC subtypes observed under a microscope
Basophils		0–2	Contain a large cytoplasm with granules and an unclear nucleus when stained.
Eosinophils		0–6	Contain a two-lobed nucleus and deep brownish red color cytoplasm, stained in a brick-red color.
Neutrophils		30–75	Contain a two-to-five lobed cytoplasm; retained pink color when stained
Lymphocytes		20–45	Contain little cytoplasm and a large staining nucleus; dense and coarse nucleus nearly the size of a normal RBC.
Monocytes		02–10	Shaped like an amoeboid; cytoplasm has no granules.

Several studies have been conducted to tackle these challenges, employing diverse machine learning (ML) models for the analysis of WBCs in microscopic images [21]. Most of the existing studies, have focused on WBC detection, segmentation, and classification [2] using traditional learning algorithms [22]. However, the challenges mentioned earlier, conventional approaches often face difficulties in achieving accurate detection and classification results for WBCs. While Deep learning (DL)-based techniques can enable the healthcare professionals to easily examine WBCs in microscopic blood cell images [2]. DL-based WBCs analysis can reduce possible human error and assist medical experts in detecting, counting, and classifying WBCs, thus enhancing the overall efficiency and accuracy of the WBCs analysis process [23]. Recently, CNN-based approaches have been extensively applied to MIA tasks, such as WBCs detection [24], semantic segmentation [16], and classification [2]. Several CNN models have been deployed with different architectures, for WBCs detection such as the R–CNN family [25], Single Short Detector (SSD) [26] and You Only Look Once (YOLO) series [24]. These detection CNNs significantly improve the performance in terms of detection; however, these models still have the problem of high computational cost because they rely on conventional sliding windows and selective search approaches.

In this study, we addressed the challenges posed by labor-intensive and error-prone manual WBC counting, segmentation, detection, and classification of microscopic images. This research is driven by the necessity for enhanced efficiency and accuracy in the analysis of WBCs within the realm of medical diagnostics. Traditional approaches often require time-consuming pre-processing steps and cell pre-segmentation, making them less than ideal for real-time healthcare applications. In the proposed framework, we used a novel WBC detection and classification approach to eliminate the need for additional preprocessing and cell pre-segmentation, thereby streamlining the diagnostic process. Our method significantly improved the accuracy and effectiveness of WBC analysis. This enhancement not only benefits medical experts in diagnosing various hematological diseases, but also positions the proposed technique as a suitable choice for resource-conscious devices in real-time healthcare systems. In the proposed methodology, we coupled dual attention (DA) with a CNN to address the issues of computational cost, model size, and detection accuracy. Moreover, the proposed WBCs detection and classification approach does not involve extra steps of preprocessing and cell pre-segmentation. The proposed method significantly improves the accuracy, effectiveness, and future real-time healthcare systems to assist medical experts in diagnosing several hematic diseases. Thus, the proposed technique can be efficiently implemented in resource-conscious devices. The contributions of our work are summarized as follows.

- In the MIA field, particularly for WBC analysis, a major challenge is the unavailability of large, well-annotated, and high-quality datasets. To address this limitation, we introduce an innovative approach that incorporates deep convolutional generative adversarial network (DCGAN) mechanism, aimed at enabling the generation of synthetic samples by utilizing knowledge from existing datasets. It enriches the

existing data to enhance system robustness and accuracy and improve the efficiency of WBCs analysis tasks.

- The proposed DA-coupled CNN-based cell analysis technique enhances the diagnostic process by supporting medical experts in diagnosing various hematic diseases. Specifically, it addresses the challenge of detecting and localizing different WBC subclasses in blood smear images. This task is difficult for traditional object detectors because of the overlapping morphological complexity and texture similarities among cells. The primary goal was to develop a novel CNN-based technique with the ability to generalize, effectively detect, and classify these challenging cells in microscopic blood cell images.
- The proposed DA-coupled CNN-based approach achieved the highest accuracy compared to existing methods, demonstrating its effectiveness and potential for advancing health diagnostic capabilities. The promising results proved that our proposed framework has the potential to significantly improve robustness, efficiency, and overall accuracy. However, thorough evaluation, validation, and integration with existing clinical workflows is essential to ensure the safety and reliability of the proposed framework in real-world healthcare diagnostic systems.

The remainder of this paper is organized as follows. Section 2 provides related literature and the limitations of existing methods used for WBCs analysis in microscopic images of blood. The proposed system, design, and detailed methodology are described in Section 3. Section 4 presents experimental results, evaluations, and detailed comparative analyses. Section 5 provides detailed discussion and limitations of the proposed method. Section 6 draws our conclusions and recommends potential areas for future research.

2. Related work

In this section, we conduct a detailed review and critical analysis of related literature, shedding light on significant improvements, methodologies employed, and challenges in precise and efficient WBCs detection and classification in blood microscopic images. By synthesizing the current state of the art (SOTA), our aim is to establish a robust understanding of the evolving techniques in this field, thereby justifying the necessity and originality of the proposed framework. In recent years, in the field of MIA and WBC classification has significant progress, propelled by the convergence of advanced imaging technologies and SOTA ML approaches. Numerous studies exited, an automating WBCs detection and classification in microscopic images, with a focus on improving diagnostic accuracy and efficiency. Prior investigations have explored diverse methodologies, spanning from traditional image processing techniques to cutting-edge DL architectures. Noteworthy contributions encompass enhancements in feature extraction, advancements in classification algorithms, and the exploration of innovative data augmentation strategies [27]. The gradual advancement of algorithms for WBC analysis can help explore their role in clinical hematology; therefore, this analysis plays a vital role in early disease diagnoses, and reduces hematological efforts by providing early and effective disease diagnostic considerations to medical experts. Advancements in ML-based MIA techniques and WBC analysis algorithms have played vital roles in clinical hematology. They aid in early disease diagnosis, reduces hematological efforts, and provides timely diagnostic insights for medical experts. In recent studies, researchers have mainly focused on the development of various DL-based WBCs analysis techniques [28]. DL-based WBCs analysis was performed to analyze the morphological shape, size, and total blood count and to classify different types of cells into their respective subtypes. The structure, size, and total count of WBCs provide an effective indication of various health disorders, such as anemia, malaria, blood cancer, leukemia, and several other hepatic diseases. Conventional techniques are typically used to perform WBCs detecting, segmenting, and classification. Conventional approaches involve three steps: WBC segmentation, feature extraction, and classification of microscopic blood images. Various conventional approaches were employed to segment, count, and classify WBCs in blood smear images. Sajjad et al. [29] integrated a K-clustering algorithm for WBC segmentation, feature extraction involving geometric and statistical aspects, and employed SVM classifiers for WBC classification in blood smear images. Safuan et al. [30] used color space correction in Lab* format, Otsu segmentation of the WBC region using RGB, CMYK, and HSV color analysis, noise reduction, and WBC counting through morphological filtering, followed by Connected Component Labelling (CCL) and Circle Hough Transform (CHT).

In WBC analysis, an effective feature descriptor directly improves the feature extraction and classification accuracy. Several conventional approaches have been deployed to extract rich features, such as HOG [31], SURF, and SIFT [32]. These descriptors use selective search and sliding-window techniques. Different pre-processing techniques, such as erosion, dilation, and color enhancement, have also been applied to facilitation the easy detection of WBCs [1]. These pre-processing techniques play a crucial role in enhancing and preserving the geometric and statistical features of WBC and in improving the accuracy of detection. Hegde et al. [33] exploited the TissueQuant approach to manage color and illumination variation enhancement techniques, and segmented WBC nuclei using thresholding. They extracted geometric features from segmented class wise nuclei to train a hybrid classifier of an SVM and ANN. The detection results illustrated that the WBCs nuclei detection was 95% with a Dice similarity coefficient. Furthermore, the classification results varied from WBCSs subtypes, that is, lymphocytes and monocytes achieved an average accuracy of 100%, and the remaining three subclasses achieved different accuracies. SVM, Bayesian, and ANN classifiers are the most commonly used methods for WBC classification in the literature. Sajjad et al. [1] extracted different geometric, color, and texture features, and an SVM achieved a high accuracy of 98.6% for classifying leukocytes. Khan et al. [2] used traditional methods to segment and classify WBCs in blood smear images. However, challenges remain, and efforts are required to improve its efficiency and accuracy.

Therefore, DL-based models are the most successful and practical approaches for effectively detecting and classifying WBC in smear images with limited resources [34]. In general, DL-based object detection algorithms are divided into two categories: 1) region-based R–CNN [35] and 2) without-region, SSD [26] and YOLO series [24]. Katar et al. [36] introduced the vision transformer (ViT) model for automated WBC detection in blood films. The model utilizes self-attention mechanisms to extract image features and was trained and validated on a public dataset of 16,633 samples containing five WBC types. The model achieved 99.40% accuracy in classifying these cells, which can significantly assist pathologists in the detection and reporting process. Kishore et al. [37] introduced a DL approach for WBC classification employing a CNN optimized through a particle swarm optimization algorithm. The method underwent testing on the combined LISC and BCCD datasets, achieving a classification accuracy of 99.2%, with 94.56% sensitivity, 98.78% specificity, and an AUC of 0.982. Among these methods, the Faster R–CNN [38] stands out as a widely employed algorithm for object detection due to its robustness, dynamism, and streamlined lightweight structure. Thus, the motivation of this study was to develop an efficient, robust, and lightweight CNN technique to detect and classify invisible and overlapping WBCs in microscopic blood images. There are various methods for detecting and classifying WBCs/leukocytes in blood smear images. Though, significant limitations connected with these strategies must be acknowledged. One such limitation is the difficulty in properly recognizing overlapping lesions, that make confusion between WBCs sub-categories for hematologist. Furthermore, conventional solutions consume a lot of resources, and lack of medical data poses a challenge for achieving precise results. In order to solve these limits, we developed a competent CNN that was integrated with both a DA mechanism and a DCGAN. This combination tries to improve precision in identifying and classifying WBCs, thereby leading to high accuracy. The advent of

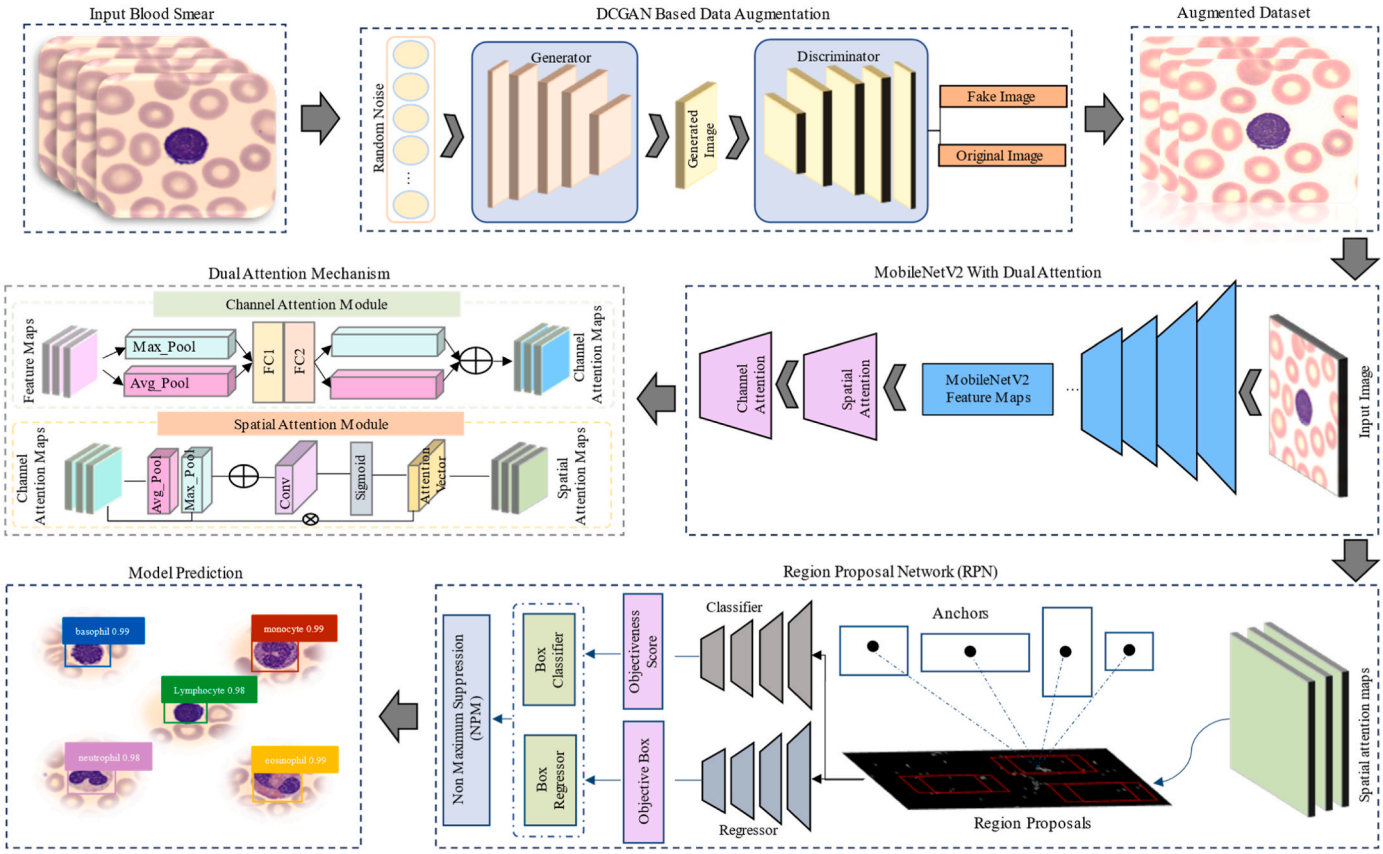


Fig. 1. The proposed WBC detection and classification framework for medical applications.

generative adversarial networks (GANs) was a turning point in the development of DL for computer vision and image processing. Out of all GAN architectures, DCGAN design has gained significant attention from researchers due to its remarkable contributions in image generation, augmentation, and enhancement. Ma et al. study [39] used GANs especially DCGAN for improving their framework that classifies blood cell images. In this regard, they creatively combined DCGAN with ResNet which is a residual neural network (ResNet) to enhance WBC classification accuracy and effectiveness in blood microscopic image. These experiments yielded promising results where an accuracy rate of 91.7% WBC classification in blood microscopic image was observed. This implies that integrating GAN technology with ResNet demonstrates how advanced DL techniques can contribute significantly towards MIA domain. In another development regarding this matter Jung et al. [40] proposed using CNNs and U-Net architecture as a DL approach for leukocyte segmentation in microscopic images. The findings underscore its edge over leading techniques. Mimosette et al. [41] introduced a DL method for segmenting WBCs in microscope images. They incorporated an attention module within convolutional blocks (CBAM) and employed DCGAN to achieve SOTA results. Barrera et al. [42] employed innovative variations of GANs and DCGAN to generate synthetic images depicting normal leukocytes and leukemic cells, addressing concerns related to class imbalance. Conversely, the training of GANs can be challenging because of several common issues, including mode collapse, vanishing gradients, and unstable training dynamics. DCGAN has been developed to increase the number of GANs, provide more robust optimization procedures, and generate more realistic synthetic images [43]. Therefore, we incorporated the DCGAN to improve the data limitations and class imbalance problems. In the proposed study, we incorporated a modified CNN model with a DA-based CNN backbone that efficiently detects and classifies WBCs into their respective sub-categories in blood microscopic cell images and achieved good results compared with SOTA methods.

3. Proposed methodology

In this section, we present a detailed discussion of the key phases of the proposed framework. It consists of two main steps, as illustrated in Fig. 1. In Step 1, we employed a DCGAN as the data augmentation mechanism, which effectively increased the sum of the training samples. In Step 2, we incorporate faster R–CNN [38], utilizing MobileNetV2 with dual attention as the backbone for the accurate detection and recognition of target objects (WBCs) in microscopic blood-cell images. The selection of MobileNetV2 as a backbone network module involved a comparison of VGG16 [44], ResNet 50 [40], ResNet 101 [39] and MobileNetV1 [41] to determine the most suitable backbone network for the proposed WBCs detection framework. MobileNetV2 with dual attention technique, is employed to extract feature maps, followed by region proposed network (RPN), to improve WBC detection and recognition accuracy (for more details see the following sections 3.1 to 3.6). Fig. 1 illustrates the network architecture of the proposed framework.

The proposed framework and algorithm, highlight the reliable considerations and methodologies employed, laying the foundation for an advanced and effective detection and classification of WBCs within microscopic blood cell images.

Proposed Algorithm.

Step 1: Pre-processing
Apply DCGAN // To generate synthetic samples from existing data
Step 2: Load Model // Fater R-CNN with backbone MobileNetV2
Step 3: Add Dual Attention Module
a. Add Channel Attention (CA) Module
Apply convolutional layer // To obtain channel descriptor
Apply global average pooling to Channel Descriptor // To obtain channel-wise Attention Vector
Apply scaling factor to the channel-wise Attention Vector
Multiply the input feature map by scaled Attention Vector // To obtain attended feature map
b. Add Spatial Attention (SA) Module
Apply convolutional layer to the input feature map // To obtain spatial descriptor
Apply global max pooling operation to the Spatial Descriptor // To obtain Spatial Attention Vector
Apply scaling factor to the Spatial Attention Vector // Match dimensions of the input feature map
Reshape the scaled Attention Vector
Multiply the input feature map by the reshaped attention vector // To obtain the attended feature map
Step 4: Train the RPN (Region Proposal Network)
Apply convolutional layer
Apply sliding window of size k x k to each feature map // To obtain feature map
Apply two separate convolutional layers // Predict object score and bounding box coordinates on each anchor box
Apply non-maximum suppression function ()
function non-maximum suppression (boxes, scores, threshold); // Remove redundant proposals, return remaining proposals
Step 5: Sort bounding boxes by their confidence scores in descending order
Apply Sorting // To sort the boxes predicted by RPN
sort_boxes_by_scores (boxes, scores) // Initialize an empty list to store picked boxes
picked_boxes = []
while boxes not empty
best_box = get_highest_score_box(boxes) // Pick the box with the highest confidence score
picked_boxes.append(best_box)
boxes = remove_overlapping_boxes(boxes, best_box, threshold) // Remove best and highly overlapping boxes from consideration
return picked_boxes
Step 6: Define ROI pooling layer and attach it to MobileNetV2
Step 7: Train MobileNetV2 to classify and detect ROIs
Step 8: Image sample predict output: // Predict WBCs sub-category

3.1. Data augmentation through conventional approaches

Data-augmentation techniques are frequently employed to address the limited availability of training images for CNNs. These techniques involve basic image manipulations, such as rotation, flipping, contrast variation, luminosity adjustments, and focal length changes. These manipulations generated additional cell samples for the learning process. The primary objective was to enrich the variety within the training data, enabling ML models to achieve better generalization and enhanced performance on unfamiliar data. Recently, novel and sophisticated methods have emerged to augment databases by synthesizing artificial images. These include advancements such as neural-style transfer [45], adversarial training [46], and utilization of GANs [47]. Nozaka et al. [48] applied three traditional data augmentation techniques, namely scaling, rotation, and distortion, to augment the quantity of data samples. However, conventional approaches constrain the variability range, leading to minimal overall alterations in the database [49]. Barrera et al. [42] proposed a GAN-based approach to enrich the data for WBCs classification. Therefore, in the proposed study, we incorporated a DCGAN to avoid data limitations and imbalance issues.

3.2. Data augmentation through deep learning

Medical images contain insufficient medical data, making it difficult to train accurate detection, segmentation, and classification models to classify WBCs in microscopic images. However, the small amount of training data limits the generalization ability and accuracy of the network [50]. Researchers have attempted different conventional and DL methods to obtain additional training data [28]. Conventional methods include cutting, moving, and turning images and adding blurry effects [51]. As traditional methods can only generate limited new data and require significant manual effort, it is important to discover more effective ways of increasing the training examples to enhance a model's performance. GANs help create more realistic synthetic images from existing images to improve the classification accuracy of models [52]. GANs serve diverse purposes including segmentation, detection, classification, creation, and reconstruction. Key models such as DCGAN [42] enhance training stability with deep convolutions, conditional GAN [53] incorporates labels for better image classification, SRGAN [54] improves high-resolution image quality, CycleGAN [55] performs domain-to-domain image translation, InfoGAN [56] focuses on learning specific features, and WGAN [57] stabilizes training using a critic instead of a discriminator in image creation. These variants address various challenges and requirements of GAN applications. Hartanto et al. [58] employed a DCGAN to generate synthetic images, aiming to balance the WBCs dataset and enhance classification accuracy. Qin et al. [59] proposed a style-based GAN architecture for skin disease detection that effectively improved the model accuracy.

Thus, recent literature reported that DCGAN is one of the best approaches for data augmentation. The DCGAN has a generator and discriminator [60]; the generator makes fake WBC samples using knowledge from real images, whereas the discriminator distinguishes real from fake; both networks train together until the difference is minimal, as shown in Fig. 2. This strategy significantly expands the dataset, increasing each WBCs subclass accordingly, resulting in a more balanced dataset that improves WBCs detection and classification performance, and ultimately enhances diagnostic capabilities in clinical hematology. In the proposed work, DCGAN data augmentation effectively expanded the existing benchmark WBCs datasets (i.e., peripheral blood cell (PBC), leukocyte images for segmentation and classification (LISC) and Raabin-WBC) by generating new samples from existing ones, as shown in Tables 3–5, respectively.

3.3. Faster R–CNN

Faster R–CNN stands out as a highly efficient object detection technique, renowned for its superior accuracy and speed when compared to earlier methods within the R–CNN family [35]. This model is an integrated object detection system comprising two modules: an RPN and a detection module [62]. The RPN module employs a deep fully convolutional network to generate proposals, and the detector module uses these proposals for object detection. In this study, we utilized the pipeline illustrated in Fig. 1 to detect and categorize WBCs in

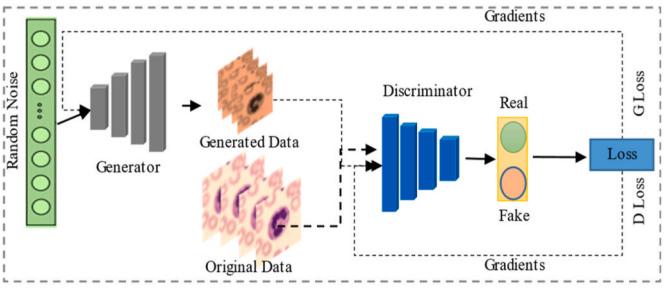


Fig. 2. DCGAN-based data augmentation.

Table 2
System hardware configurations and software specifications for the proposed technique.

Name of Hardware	Configuration	Model/Version
System	Core i5 10th Generation	B560 M AORUS PRO AX
GPU	NVIDIA	GeForce RTX-3060
Operating System	Windows 10	Pro 64-Bit
Web Platform	Jupyter Notebook	–
IDE	Anaconda	–
Programing Language	Python	Version 3.9
Imaging Libraries	OpenCV 4.5.0	Version 4.5.0

Table 3
WBC sub-category details of the PBC dataset before and after data augmentation.

Dataset	WBC type	Before Augmentation	After Augmentation
PBC	Neutrophils	3329	3561
	Eosinophils	3117	3896
	Basophils	1218	3522
	Lymphocytes	1214	3517
	Monocytes	1420	3775
	Grand Total	**10298**	**182711**

Table 4
WBC sub-category details of the LISC dataset before and after data augmentation.

Dataset	WBC type	Before Augmentation	After Augmentation
LISC	Lymphocytes	100	1000
	Monocytes	100	1000
	Neutrophils	100	1000
	Eosinophils	100	1000
	Basophils	100	1000
	Grand Total	**500**	**5000**

Table 5
WBC sub-category details of the Raabin-WBC dataset before and after data augmentation.

Dataset	WBC type	Before Augmentation	After Augmentation
Raabin-WBC	Lymphocytes	3461	3461
	Monocytes	795	3180
	Neutrophils	8891	8891
	Eosinophils	1066	3198
	Basophils	301	3010
	Grand Total	**14,514**	**21,740**

microscopic images. This approach involves the use of a CNN backbone with an attention module to extract feature maps. These maps were then processed by an RPN [63] to estimate the bounding boxes for regions that may contain target WBCs. Subsequently, the region of interest (ROI) pooling layer merges features maps with bounding box proposals, allowing the classifier to output WBC classes and the regressor to predict the bounding box coordinates. The proposed technique employs ROI pooling to detect objects within a given blood smear image. This step involves merging the feature maps extracted by the proposed dual-attention CNN and the proposals generated by the RPN into a unified vector using an adaptive pooling layer. After processing the combined inputs by two fully connected layers, the output is supplied to the regressor and Softmax, which predict the bounding box and class label of the object, respectively. The entire system is trained as a single network with shared features. The RPN guides the detector module where to look and utilizes a Selective Search method to determine region proposals [63].

3.4. MobileNetV2 architecture

MobileNetV2 introduces an improved version with an inverted residual structure and removes nonlinearities from narrow layers [64]. For efficient and effective object detection and classification MobileNetV2 achieved SOTA performance. In this model, a depthwise separable convolution initially applies a single filter to each input channel, followed by pointwise convolution to merge the output. This separation of the standard convolution operation enables the model to execute the same computation with reduced parameters, thereby making it more efficient. Fig. 3 illustrates a comparison between standard convolutions, depthwise separable convolutions, and pointwise convolutions. The mathematical representations of the standard convolution, depth-wise separable, and 1 × 1 convolutions and their computation costs are shown as follows in Eqs. (1)–(6), respectively.

$$G_{k,l,n} = \sum_{i,j} K_{i,j,m,n} \cdot F_{k+i-1,l+j-1,m} \tag{1}$$

$$D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F \tag{2}$$

Where K is the filter of size i, j, m, n, and F is the input feature map with size $D_F \times D_F \times M$, where M is the depth and D_F, represents the spatial width and height, and produces an output map G with size $D_F \times D_F \times N$ with depth N and spatial width and height D_F.

$$D_G \cdot G_{k,l,m} = \sum_{i,j} K_{i,j,m} \cdot F_{k+i-1,l+j-1,m} \tag{3}$$

In standard convolutions, the convolution kernel K of size $D_K \times D_K \times M \times N$, the spatial dimension of the kernel assumed to be square is D_K and the number of input and output channels represented are as M and N. K represents the filter of size $D_K \times D_K \times M$ where m_{th} kernel in K is applied to the m_{th} channel in F to produce the m_{th} of the kernel output feature map G.

$$D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F \tag{4}$$

In Eq. (4), we defined the total computational complexity of the neural network. This equation combines two fundamental components to provide a comprehensive overview of the computational and parametric requirements of the network. The first part of the equation ($D_K \times D_K \times M \times D_F \times D_F$) quantifies the computational and parameter loads associated with the convolutional layers. $D_K \times D_K$ represents the spatial dimensions of our convolution kernels, which are typically square with a side length of D_K. M denotes the number of input channels. $D_F \times D_F$ signifies the spatial dimensions of the feature maps. This equation precisely accounts for the computational and parameter demands of the convolutional layers.

The second part of equation ($M \times N \times D_F \times D_F$) encapsulates additional operations or parameters that are often linked to fully connected or other layers within the neural network. To elucidate this further, M represents the number of input channels, N denotes the number of output channels, and $D_F \times D_F$ denotes the spatial dimensions of the feature maps.

$$\frac{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F + M \cdot N \cdot D_F \cdot D_F}{D_K \cdot D_K \cdot M \cdot D_F \cdot D_F} = \frac{1}{N} + \frac{1}{D_k^2} \tag{5}$$

In the proposed method, we used MobileNetV2 as the backbone for WBC detection with Faster R–CNN, and a dual-attention mechanism was included to improve feature representation and grasp crucial context in blood smear images to recognize complex and overlapping WBCs in blood microscopic images. Compared with SOTA techniques, the empirical results demonstrate the superior performance of the proposed model.

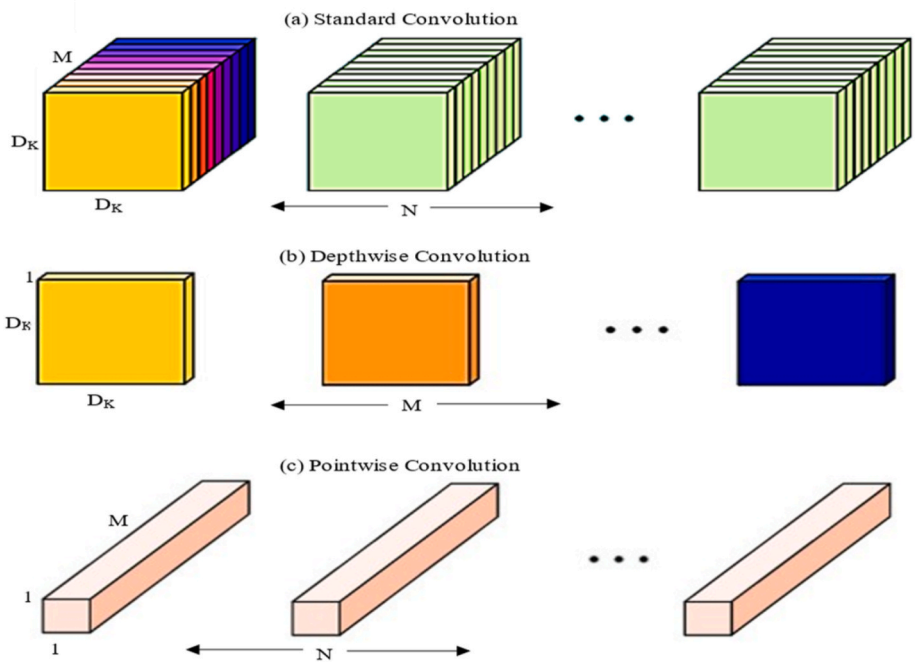


Fig. 3. Comparison of (a) Standard, (b) depthwise separable and (c) pointwise convolutions [61].

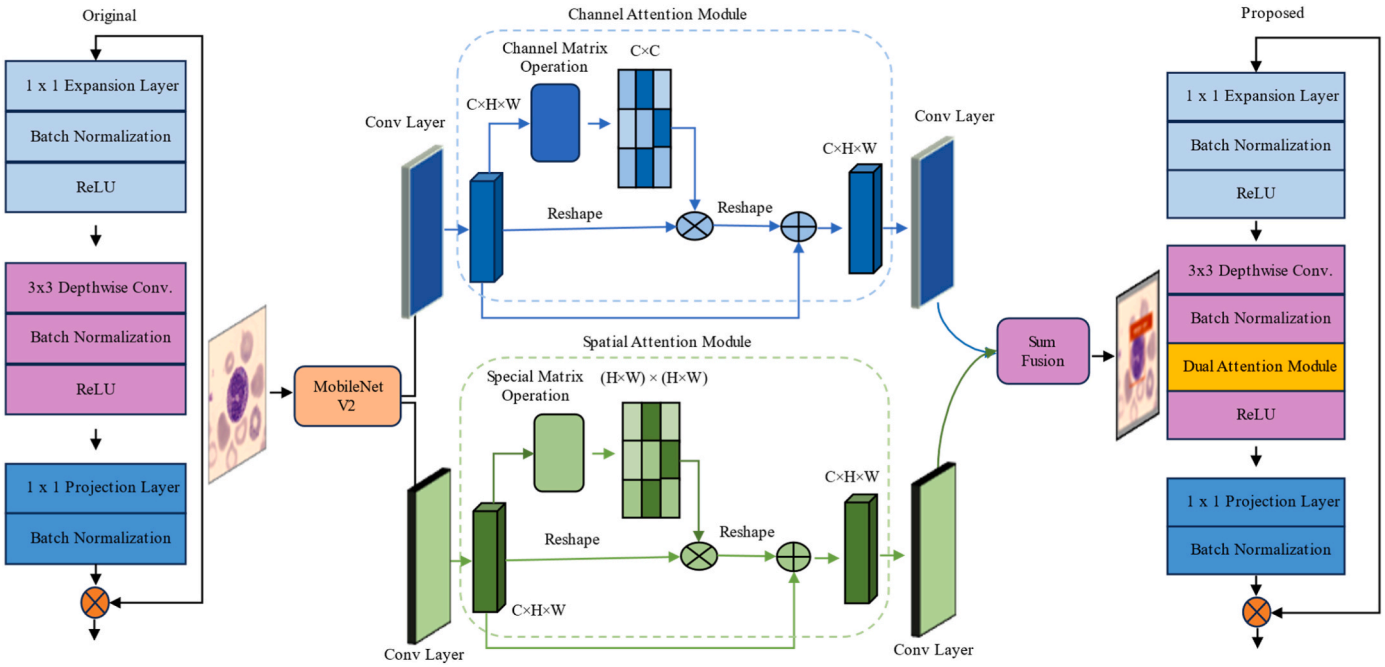


Fig. 4. The overall view of our proposed feature extraction backbone with Dual Attention Module.

3.5. Dual attention approach

In different fields, researchers have tried CNN-based structures with attention methods, particularly for videos because of their visual similarities [24]. However, for images, channel attention (CA) or spatial attention (SA) methods have had limited success because of the size and dimensions of the data [65]. Certain studies have used CA or SA modules in CNN architectures to detect objects in images. Currently, attention mechanisms have been applied across different domains in different ways [66]. In the proposed method, we use the DA mechanism to detect and classify WBCs in blood smear images by capturing high-level and fine-grained details to enhance the detection and classification accuracy. An overview of the proposed DA Module is presented in Fig. 4.

a) **CA module:** This method enhances model accuracy by paying more attention to important channels of feature maps, while disregarding less relevant ones. By doing so, the feature quality is improved, boosting the model performance with a minor computational impact. In our study, we utilized CA to capture the relationships between feature map channels using convolution, global average pooling, maximum pooling, and fully connected layers. During training, different feature map channels contribute to object representation, and our approach generates two background descriptions, F ăCavg and ăCmax, providing background knowledge about the feature map.

$$ăCavg = Avg\text{-}Pool(ă), ăCmax = Max\text{-}Pool(ă) \quad (6)$$

First, the output descriptions of ăCavg and ăCmax are received by two shared FC layers, fc1 and fc2. Then, the ReLU activation function was applied to each layer to obtain Mavg and Mmax. After that, the weight coefficients are obtained by adding the two feature maps together, resulting in Mć(ă).

$$\acute{M}max = \ominus (f\ c2(\ominus (f\ c1\ (\breve{a}\ Cmax)))), \acute{M}avg = \ominus (fc2\ (\ominus (f\ c1\breve{a}Cavg)) \tag{7}$$

$$\acute{M}ć(\breve{a}) = (\acute{M}max \oplus \acute{M}avg),\ Fc = \acute{M}ć(\breve{a}) \otimes \alpha \tag{8}$$

The fc1 and fc2 layers apply 1 × 1 convolution for compress and decompress features of every channel, with addition represented by ⊕, and activation function ReLU (represented by ⊖), as shown above in Fig. 4. Ḿć(ă) features are obtained by adding Mmax and Mavg, followed by introducing a skip connection (represented by ⊗) to preserve the input feature flow and obtain the CA map Fc.

b) **SA Module:** This module pinpoints crucial areas within the image. While the SA module identifies relationships among different image regions, the CA module identifies key aspects across multiple channels. SA employs average pooling and max pooling along the channel, merging them for effective feature descriptions and highlighting informative image regions. Unlike the CA module, SA emphasizes finding vital image regions and extracting the spatial relationships between features. In the SA module, it takes an input feature map F and creates two intermediate feature maps using two pooling operations, ăCavg and ăCmax, both of which have dimensions 1xHxW.

$$\breve{a}Savg = \text{Avg-Pool (Fc)},\ \breve{a}Smax = \text{max-Pool (Fc)} \tag{9}$$

Subsequently, a 2D feature map was formed with SA, and the feature maps were combined and convolved using three convolutional layers, each with 64 filters and ReLU activation. The layer sizes are 1 × 1, 3 × 3, and 1 × 1. We opted for standard convolution instead of dilated convolution [67].

$$\acute{M}sFc = (\ominus(f1 \times 1\ (\ominus(f3 \times 3(\ominus(f1 \times 1(\breve{a}Savg \oplus \breve{a}Smax)))))), \tag{10}$$

Convolutional layers in SA use filter size f. For the SA map ḾsFcGAP, global average pooling is applied to ḾsFc, then concatenated with Fc for a combined map is denoted as follows:

$$\acute{M}sF_{cGAP} = G\ AP(Fc),\ Fs = \odot\ [\acute{M}sF_{cGAP}, Fc] \tag{11}$$

We get Fs by applying concatenation, then batch normalization is employed to Fs, and the resulting feature maps are concatenated with ă to acquire FCS.

$$FCS = \odot\ [\text{Batchnorm}, \breve{a}] \tag{12}$$

3.6. Region proposal network (RPN)

An RPN is an end-to-end fully convolutional network that predicts the object boundaries and scores for each detection during training. In 2015, Ren et al. [63] claimed that this algorithm was robust against scale and aspect ratios. In the proposed method, feature maps are extracted using an efficient CNN coupled with a DA backbone, and the RPN is then used to estimate the bounding boxes (also known as proposed regions) to identify regions that may contain the target object, as shown in Fig. 5. In the proposed approach, the RPN employs a 3 × 3 window that traverses a high-level convolutional feature map. This process yields an effective window size of 256 × 256 when projected back onto the input layer. This allows the RPN to use many contexts to generate proposals. The 3 × 3 window was resampled into a 256-dimensional vector to generate proposals and then fed into two fully connected layers.

The regressor layer had 4k outputs (x, y, w, and h), whereas the classification layer had 2k outputs (background and foreground), resulting in 4k + 2k outputs per position. The bounding box regression head outputs the bounding boxes for each anchor box at each location in the convolution layer. The classification layer produced confidence scores that indicated the presence of an object within each anchor box at each location in the convolution layer. The criteria used by these anchors to assign labels were based on two things: 1) the ground truth box and the anchors with the highest intersection-over-union overlap, and 2) the intersection-over-union anchors are greater than the threshold required to identify objects in the input image.

4. Experimental results

In this section, we briefly explain the system configuration, datasets, and different evaluation parameters. Several experiments have been conducted, such as conventional data augmentation, with DCGAN data augmentation, with attention, and without an attention module. The proposed model was evaluated through these experiments during training and validation, followed by a comparative analysis with SOTA models. Therefore, the proposed approach is superior to existing models for classifying WBCs in microscopic blood cell images in terms of accuracy, sensitivity and specificity.

4.1. System configuration

The proposed framework was evaluated on the system with hardware configuration and software specification details listed in Table 2. A 10th generation Core i5 (2.9 GHz) CPU with 16 GB of RAM and a 12 GB NVIDIA GeForce RTX-3060 GPU was used. Jupyter Notebook as a web platform and the user-friendly IDE Anaconda were utilized during the evaluation process. The Python 3.9 programming language is utilized with several image processing libraries such as OpenCV version 4.5.0, and Scikit-Learn.

4.2. Datasets explanation

In this subsection, we provide the details of the datasets which were utilized in our proposed study. The performance of the proposed technique is evaluated using three benchmark datasets: PBC, LISC, and Raabin-WBC. Prior to training, we implemented data augmentation to enhance the existing data, aiming to improve model accuracy and generalization. Data augmentation was applied to each subclass of WBCs under various conditions, with a specific emphasis on augmenting data for subclasses with fewer samples compared to those with the maximum number of samples. Tables 4–6 showcase both the original and augmented datasets. Subsequent to the application of data augmentation, each dataset was partitioned into three folders representing

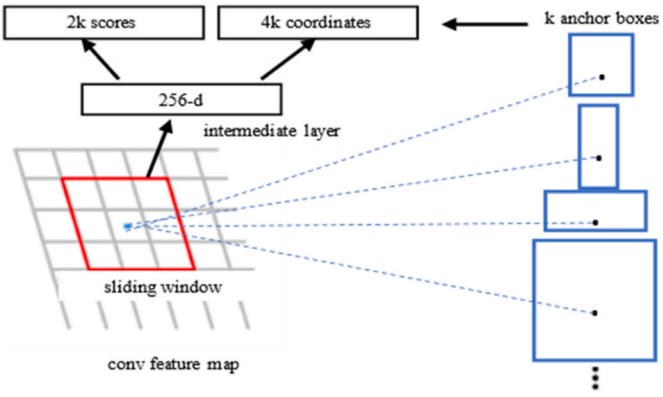


Fig. 5. RPN generated Anchor boxes [68].

Table 6
Confusion matrix for PBC dataset with Conventional data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.95**	0.00	0.02	0.00	0.02
Monocyte	0.00	**0.96**	0.02	0.02	0.00
Eosinophil	0.01	0.00	**0.89**	0.05	0.04
Neutrophil	0.03	0.00	0.03	**0.94**	0.01
Basophil	0.03	0.00	0.00	0.04	**0.93**
Overall Accuracy (%)			**93.83**		

training, validation, and test sets for assessing the proposed framework performance. During the training phase, 70% of the images from the dataset were utilized, with 20% allocated for validation, and the remaining 10% reserved for testing.

(a) **PBC-WBC Dataset;** The PBC dataset, originating from the Core Lab at the Hospital Clinic of Barcelona, Spain, is accessible online via Mendeley Data. This dataset was curated for the development of automatic recognition algorithms [69]. It comprises 17,092 microscopic images depicting healthy cells, categorized into eight subcategories based on size, color, and structure. These subcategories include neutrophils, eosinophils, basophils, lymphocytes, monocytes, immature granulocytes, erythroblasts, platelets, and thrombocytes. Captured at a resolution of 360 × 360 in JPG format, the images were annotated by hematologists. In our study, we utilized a modified version of the PBC dataset, specifically selecting the WBCs classes, namely neutrophils, basophils, eosinophils, monocytes, and lymphocytes. Our focus centered exclusively on the detection and classification of WBCs. The modified PBC dataset comprises 10,298 samples of WBC subclasses, with each subcategory containing a varying number of images, as detailed in Table 3.

(b) **LISC Dataset;** The LISC database consists of hematological images sourced from the peripheral blood of individuals without any health issues. These images were compiled at the BMT Research Center of Hematology and Oncology, Imam Khomeini Hospital in Tehran, Iran [70]. A hematologist categorized the blood cells into regular leukocytes, encompassing basophils, eosinophils, lymphocytes, monocytes, and neutrophils. Additionally, an expert manually outlined the regions corresponding to the nucleus and cytoplasm in the images. Originally, the dataset included samples obtained from eight healthy subjects, resulting in a total of 400 samples collected from 100 microscope slides. Each slide contained more than one WBC subclass image. We manually extracted and stored 100 images from the slides of each WBC subclass, totaling 500 images, as depicted in Table 4.

(c) **Raabin-WBC Dataset;** The Raabin-WBC dataset [71] consists of 14,515 microscopic images of the following WBCs subclasses: lymphocytes, monocytes, neutrophils, eosinophils, and basophils, with a count 3461, 1420, 8891, 1066, 301 respectively as shown in Table 5. In this dataset, each subclass of WBC cell images was saved in JPG format with a resolution of 575 × 575 pixels and ground truth labels were provided by two hematology experts for each image [57].

4.3. Model evaluation parameters

In this section, we explain several evaluation parameters used to validate the proposed technique. These evaluation metrics include the bounding box loss, object loss, regression loss, class loss, recall, specificity, precision, mAP, and accuracy. These evaluations provided valuable insights into the effectiveness of the proposed model for detecting and classifying WBCs/leukocytes in blood smear images. The box loss evaluates the accuracy of the model in predicting the location and size of the bounding boxes around the detected objects. It is calculated as the average difference between the predicted and ground-truth bounding boxes. A lower box loss indicates that the model is more efficient at predicting the correct bounding boxes with higher accuracy.

$$\text{Box Loss} = \lambda_\text{loc} * \text{Localization Loss} + \lambda_\text{obj} * \text{Objectness Loss} \quad (13)$$

Here, in Eq. (13), λ_loc and λ_obj represent the weights assigned to the localization loss and objectness loss, respectively. These weights can be adjusted according to the relative importance of each component in the overall loss function. The object loss was determined by evaluating the average difference between the predicted object bounding boxes and ground truth bounding boxes in the training set. A lower object loss indicated that the model accurately detected WBCs in blood microscopic images.

$$\text{Objectness-Loss} = [y_\text{true} * \log(y_\text{pred}) + (1-y_\text{true}) * \log(1-y_\text{pred})] \quad (14)$$

In Eq. (14), y_true represents the ground truth objectness label (1 for the presence of an object, 0 for its absence), and y_pred represents the predicted objectness score.

Regression loss was employed to calculate the difference in the bounding box coordinates between the predicted and actual bounding boxes.

$$\text{Regression Loss} = \sum \text{smooth_L1}(\Delta x), \text{smooth_L1}(\Delta y), \text{smooth_L1}(\Delta w), \text{smooth_L1}(\Delta h) \quad (15)$$

$$\text{Smooth_L1}(x) = 0.5 * x\text{^}2 \text{ if } |x| < 1 \; |x| - 0.5 \text{ otherwise} \quad (16)$$

Here, in Eqs. (15, and 16) Δx, Δy, Δw, and Δh represent the differences between the predicted box coordinates and the ground truth box coordinates. Classification loss is a measure of a model's performance in classifying the correct labels of detected objects in microscopic blood cell images. The classification loss was computed by measuring the average difference between the ground-truth class labels and the model-predicted class labels. A lower classification loss indicates that the model demonstrates improved accuracy in correctly classifying the detected WBCs in the blood smear images.

$$\text{Classification Loss} = -\sum [y_\text{true} * \log(y_\text{pred})] \quad (17)$$

In Eq. (17), y_true are the ground truth class labels, and y_pred are the predicted class probabilities.

Recall/sensitivity: It is the ratio of correctly identified true positive (TP) instances to the total number of actual positive instances. It measures the accuracy of the model in recognizing positive cases among all instances that are truly positive.

$$\text{Recall or Sensitivity} = TP / (TP + FN) \quad (18)$$

Specificity: It evaluates how well a model can distinguish between the total number of true negative (TN) instances contained in the dataset and false negative (FN) occurrences such as non-target items. A higher specificity indicates the effectiveness of the model in avoiding false positives (FP) and accurately identifying negative instances.

$$\text{Specificity} = TN / (TN + FP) \quad (19)$$

Precision: The precision quantifies the percentage of properly

Table 7
Confusion matrix for LISC dataset with conventional data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.88**	0.00	0.03	0.05	0.04
Monocyte	0.00	**0.86**	0.03	0.06	0.05
Eosinophil	0.03	0.01	**0.89**	0.04	0.03
Neutrophil	0.04	0.02	0.05	**0.85**	0.04
Basophil	0.04	0.01	0.02	0.03	**0.90**
Overall Accuracy (%)			**87.50**		

Table 8
Confusion matrix for Raabin-WBC dataset with conventional data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.93**	0.03	0.03	0.00	0.04
Monocyte	0.02	**0.92**	0.03	0.00	0.03
Eosinophil	0.01	0.00	**0.95**	0.02	0.02
Neutrophil	0.001	0.00	0.03	**0.90**	0.06
Basophil	0.03	0.0	0.03	0.00	**0.94**
Overall Accuracy (%)			**92.80**		

detected positive examples (such as items of interest) among all instances projected as positive.

$$\text{Precision} = \text{TP}/\ (\text{TP} + \text{FP}) \tag{20}$$

mAP assesses the model's accuracy for various recall levels for a given class.

$$\text{mAP} = (1\ /\ \text{N})\ *\ \sum\ (\text{Recall_i}\ *\ \text{Precision_i}) \tag{21}$$

Accuracy: The proportion of accurate forecasts to all predictions made is calculated to assess the general accuracy of the model's predictions.

$$\text{Accuracy} = (\text{TP} + \text{TN})\ /\ (\text{TP} + \text{TN} + \text{FP} + \text{FN}). \tag{22}$$

4.4. Results with conventional data augmentation

The confusion matrices for the PBC, LISC, and Raabin-WBC datasets using conventional data augmentation techniques (for details, see Section 3.1) are listed in Tables 6–8, respectively. Table 6 reveals that the accuracy values for the PBC dataset were 0.95, 0.96, 0.89, 0.94, and 0.93 corresponding to each WBCs sub-class (lymphocyte, monocyte, eosinophil, neutrophil, and basophil), and the cumulative accuracy was 93.83%. In Table 7 the accuracies achieved for WBCs subclasses on LISC dataset are 0.88, 0.86, 0.89, 0.85 and 0.90 respectively and with overall accuracy is 87.50%; likewise, from Table 8 the accuracies are included 0.93, 0.92, 0.95, 0.90 and 0.94 for each WBC sub-classes (i.e., lymphocyte, monocyte, eosinophil, neutrophil and basophil) respectively with overall accuracy of 92.80% on Raabin-WBC dataset. While the outcomes may not be entirely compelling, they can be relied upon for practical applications in the healthcare field. Some WBCs were misclassified because of structural similarities between subclasses; for example, in Table 6, the lymphocyte subclass was 0.02% confused with basophils and 0.02% confused with eosinophils. The monocyte subclass was 0.02% with eosinophils and 0.02% with neutrophil subclasses of WBCs. Moreover, 0.09% of the eosinophils were confused with the other three subclasses (i.e., 0.05% confused with neutrophils, 0.04% confused with basophils, and 0.01% confused with lymphocytes), 0.03% confused with eosinophils and 0.01% confused with lymphocytes.

Similarly, 0.06% of the basophil class was confused with eosinophils and neutrophils. Additionally, Table 7 presents the confusion matrices for the LISC and Raabin-WBC datasets, incorporating our proposed system with conventional data augmentation, along with the corresponding overall accuracies. In Table 8, the overall accuracies on the LISC and Raabin-WBC datasets are 87.50% and 92.80%, respectively.

4.5. Results with DCGAN data augmentation

In this section, the performance of the proposed system is evaluated after implementing DCGAN data augmentation (across all the datasets (i.e., PBC, LISC, and Raabin-WBC). The results obtained after applying the DCGAN data augmentation are compelling, substantiating their relevance in real-world scenarios. The confusion matrix for the PBC dataset, incorporating DCGAN data augmentation, is provided in Table 9. Noteworthy improvements include the accuracy for the WBCs subclass lymphocytes, which rose from 0.95 to 0.98. Similarly, the accuracies for monocytes, eosinophils, neutrophils, and basophils exhibited enhancements of 0.96 to 0.99, 0.89 to 0.97, 0.94 to 0.98 and 0.93 to 0.96, respectively. The overall accuracy increased substantially, reaching 97.60, surpassing the accuracy achieved using conventional data augmentation approaches. Similarly, Tables 10 and 11 show that the application of data augmentation resulted in a significant improvement, with an overall accuracy increase of 3.2% for the LISC and 3% for the Raabin-WBC datasets, respectively. These findings underscore the beneficial effect of DCGAN data augmentation on the overall accuracy of the proposed technique. The proposed technique achieved average accuracies of 90.60% and 95% for LISC, and Raabin-WBC dataset respectively. These results emphasize the effectiveness of DCGAN-based augmentation in improving the model performance across diverse WBCs

Table 9
Confusion matrix for PBC dataset with DCGAN data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.98**	0.00	0.02	0.00	0.00
Monocyte	0.00	**0.99**	0.00	0.00	0.01
Eosinophil	0.00	0.00	**0.97**	0.00	0.00
Neutrophil	0.00	0.00	0.01	**0.98**	0.01
Basophil	0.02	0.01	0.01	0.00	**0.96**
Overall Accuracy (%)			**97.60**		

Table 10
Confusion matrix for LISC dataset with DCGAN data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.90**	0.00	0.02	0.05	0.03
Monocyte	0.02	**0.91**	0.00	0.03	0.04
Eosinophil	0.02	0.00	**0.89**	0.04	0.05
Neutrophil	0.01	0.01	0.03	**0.90**	0.05
Basophil	0.04	0.00	0.04	0.03	**0.91**
Overall Accuracy (%)			**90.60**		

Table 11
Confusion matrix for Raabin-WBC dataset with DCGAN data augmentation.

WBCs	Lymphocyte (%)	Monocyte (%)	Eosinophil (%)	Neutrophil (%)	Basophil (%)
Lymphocyte	**0.94**	0.00	0.01	0.02	0.03
Monocyte	0.00	**0.95**	0.02	0.01	0.02
Eosinophil	0.00	0.00	**0.97**	0.02	0.01
Neutrophil	0.00	0.01	0.02	**0.94**	0.03
Basophil	0.02	0.00	0.01	0.03	**0.95**
Overall Accuracy (%)			**95.00**		

classification, contributing to a more accurate and reliable classification system for MIA.

Fig. 6 illustrates the visualization of training time and validation accuracies for each epoch, comparing conventional and DCGAN data augmentation techniques with the PBC dataset. It highlights the noticeable improvement in the model accuracy. This observation underscores the importance of obtaining a significant amount of data for the successful deployment of DL techniques. Specifically, the accuracy of the first epoch increased from 60% with the conventional data augmentation to 66% with the DCGAN data augmentation approach.

4.6. Comparative evaluation of the proposed DCGAN with SOTA

In this section, we evaluate the efficacy of the proposed method compared to SOTA enhancement techniques. During the evaluation, the proposed technique was individually applied to each WBC subclass, and the outcomes were compared with SOTA enhancement techniques. Table 12 provides a comparison in terms of sensitivity, specificity, and accuracy with current techniques using the PBC dataset. Debapriya et al. [72], introduced a hybrid GAN architecture designed to generate synthetic images of cell types and achieved an accuracy of 95.66% on this dataset. Hazra et al. [73] proposed a GAN model incorporating the loss function of a Wasserstein GAN with a gradient penalty and achieved runner-up sensitivity, specificity, and accuracy of 97%, 97.1%, and 96.98%, respectively. Similarly, Barrera et al. [42] obtained 96.71%, 95.80%, and 95.76% accuracy, sensitivity, and specificity, respectively.

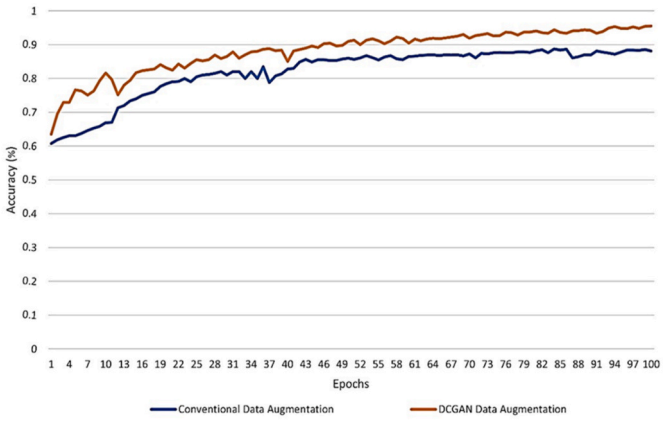


Fig. 6. Validation accuracy with Conventional and DCGAN data augmentation on PBC test data.

Bairaboina et al. [74] employed a novel GAN approach to synthesize images depicting leukocytes and leukemic cells, with 2_{nd} runner up specificity of 95.45%, sensitivity of 96.76%, and accuracy of 96.86%. Our proposed method outperformed current techniques, attaining the highest values of 97.32%, 97.44%, and 97.60% for specificity, sensitivity, and accuracy, respectively. These results represent the most superior performance achieved to date on the PBC dataset.

4.7. Results before dual attention

In this section, we evaluate the efficacy of the proposed framework for the detection and classification of WBCs prior to integrating an attention mechanism. A comprehensive evaluation was undertaken to demonstrate the robustness and performance of the proposed technique using the PBC dataset. Throughout the training phase, various parameters were meticulously examined, including box loss, object loss, class loss, and regression loss, as illustrated in Fig. 7. The values of a few parameters were very high, and some were too low, contrary to our expectations, during the training process. It was specifically observed that the regression loss did not show the expected considerable reduction. In contrast, the object loss, regression loss, and classification loss all showed considerable reductions.

This observation led a more in-depth examination of the training dynamics. Despite our predictions for a significant reduction in regression loss, the observed behavior raises interesting concerns about the relationship between different components of the model during training. Notably, while the object loss, regression loss, and classification loss all showed positive trends, the relative behavior of the regression loss encouraged more research into potential optimization options. In conclusion, our analysis of the PBC dataset not only confirms the general effectiveness of our WBC detection and classification approach, but also provides useful insights into the subtle dynamics of individual training parameters. Moreover, it establishes the foundation for a deeper understanding of the models learning process and identifies potential opportunities for optimization and improvement.

Table 12
Comparison of the proposed DCGAN with SOTA on PBC dataset.

Method	Specificity (%)	Sensitivity (%)	Accuracy (%)
Debapriya et al. [72]	95.50	94.91	95.66
Hazra et al. [73]	97.00	97.1	96.98
Barrera et al. [42]	95.76	95.80	96.71
Bairaboina et al. [74]	95.45	96.76	96.86
Proposed GCGAN	**97.32**	**97.44**	**97.60**

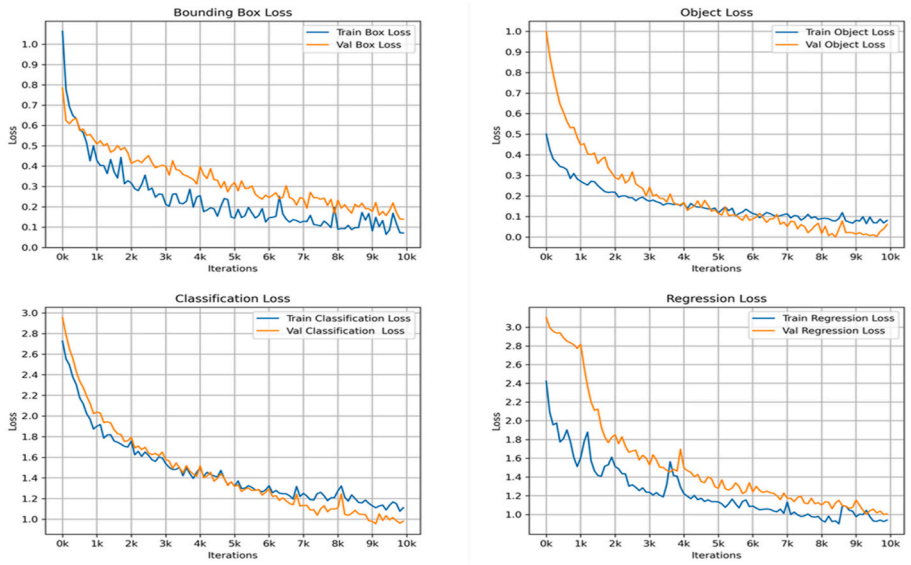


Fig. 7. Before DA results in terms of box loss, object loss, regression loss, and classification loss.

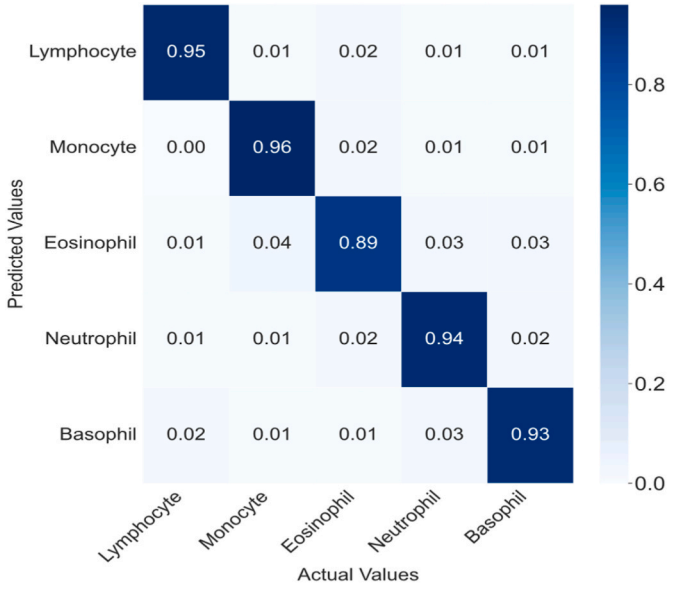


Fig. 8. Confusion matrix before DA.

This suggests that the proposed technique may have an impact on detection confidence, but the overall findings are good when compared to SOTA procedures, but not trustworthy enough to be applied in a real-world clinical setting. These results provide valuable information for further improving the performance of the proposed technique. Fig. 8 provides the confusion matrix of the proposed technique, and the achieved accuracies of an individual WBCs subclass on the PBC dataset before DA are 0.95, 0.96, 0.89, 0.94, and 0.93 for lymphocyte, monocytes, eosinophils, neutrophils, and basophils, respectively. However, while these results are not noteworthy, they do not inspire complete confidence because of the presence of confusion among the WBCs subclasses. The structural and textural similarities between classes, such as basophils, neutrophils, and eosinophils, lead to misclassification; similarly, the monocyte class exhibits a degree of structural similarity with both the basophil and eosinophil classes, while the neutrophil class has close structural resemblances with the monocyte and eosinophil classes. The model achieved an overall accuracy of 93.84% on the PBC dataset, which requires further improvement. Prior to incorporating the attention mechanism, our technique faced challenges in accurately detecting and classifying subcategories of WBCs in microscopic images. Color, morphological, and textural similarities among WBCs sub-classes that affect detection and classification effectiveness in real-world settings. To overcome these constraints and improve system performance, additional modifications, such as the incorporation of a DA module, may be required. In the proposed technique, we applied an effective DA approach to address this issue.

4.8. *Results after applying dual attention*

This part focuses on evaluating our proposed model from several perspectives, which includes an attention mechanism for detecting and classifying WBCs in blood smear images. Fig. 9 depicts an evaluation of the proposed model during training and validation. The plots show how the bounding box, object, regression, and classification losses have changed. The blue line in the shown graphs shows training performance, whereas the orange line represents validation of each evaluation parameter. After incorporating the DA module into the proposed technique, there was a significant improvement in the performance, leading to promising results. The performance of the proposed technique was also assessed by considering the confusion matrix, as given in, during training and validation, and it significantly reduced the box loss, object loss, class loss, and regression loss. The evaluation of the proposed model after incorporating the DA mechanism revealed its exceptional performance in the detection and classification of WBCs. The confusion matrix in Fig. 10 shows that the proposed detection model exhibited convincing accuracy for each WBC subclass. The model accuracies include 1.00, 0.99,1.00,1.00 and 0.99, for lymphocytes, monocytes, eosinophils, neutrophils, and basophils, respectively, on the PBC dataset. Additionally, there were significant improvements in the average accuracy of 99.83 %, 99.35 %, and 99.60% for the PBC, LISC, and Raabin-WBC benchmark datasets, respectively. The results demonstrate the superiority of the proposed method over existing techniques.

4.9. *Comparison of the proposed technique with SOTA*

In this section, we present a comprehensive comparison of the proposed framework with SOTA techniques. In order, to assess the effectiveness of the proposed model, we utilized the PBC [75], LISC [70], and Raabin-WBC [71] benchmark datasets. The evaluation process included a comparative analysis of various other SOTA, as outlined in Table 13.

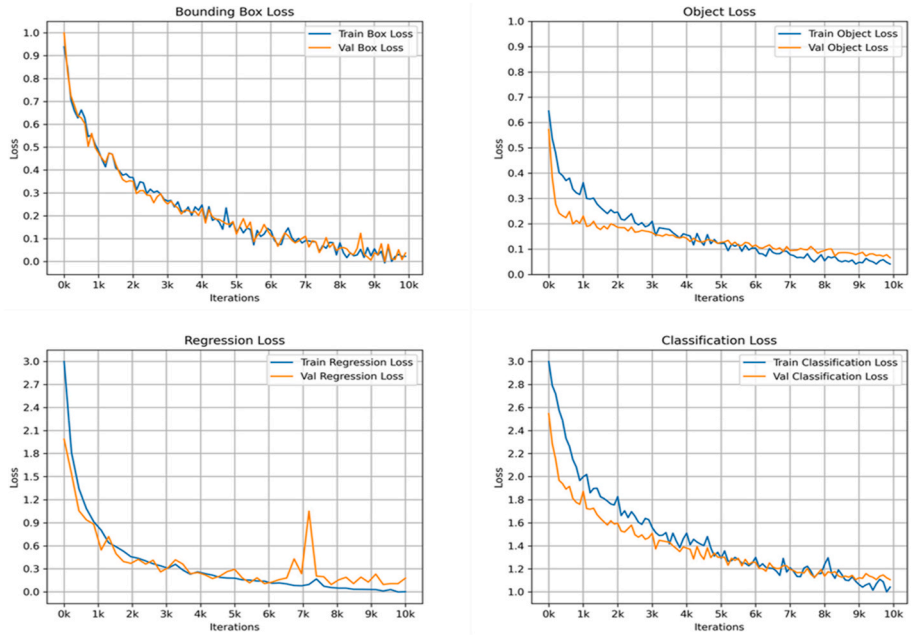


Fig. 9. After DA results in terms of box loss, object loss, regression loss, and classification loss.

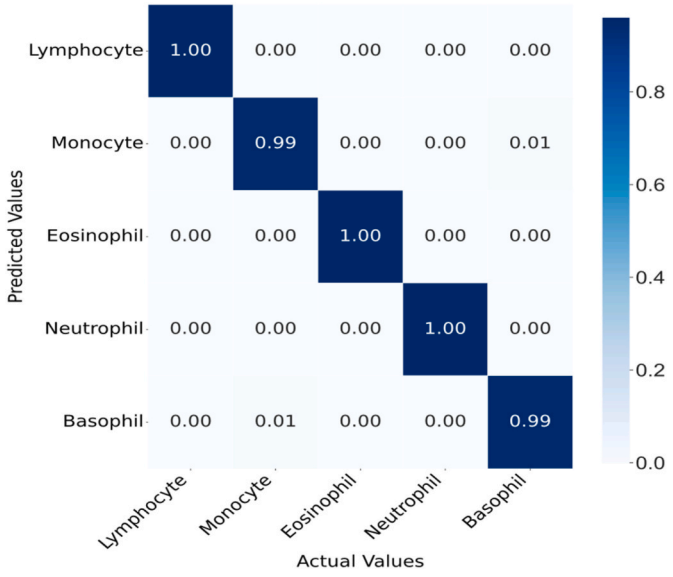


Fig. 10. Confusion matrix after DA.

This comparative analysis enables the researchers to evaluate the strengths and weaknesses of the proposed technique. It helps them to determine how the technique ranks in terms of overall accuracy compared to other existing approaches. For instance, Zhang et al. [76] employed an average ensemble model to enhance the classification performance with a testing accuracy of 90.1% using the PBC Dataset. Wang et al. [77] adopted an SSD and an improved version of YOLO for WBCs recognition. Similar results are achieved by other SOTA methods, that are surpassed by our approach.

4.10. Visual results of the proposed model on benchmark datasets

In this section, we assess the generalization ability of the proposed " efficient DA-couple with a CNN" approach. This evaluation encompasses the utilization of three well-recognized benchmark datasets: PBC [75], LISC [70], and Raabin-WBC [71]. In the qualitative analysis, we test image from each WBCs sub-class ("lymphocyte," "monocyte," "basophil," "eosinophil," and "neutrophil") from these datasets and examine visually. The goal of this qualitative analysis was to evaluate the generalization performance of the proposed detection technique on these three benchmark datasets. Fig. 11 shows the visual outcomes and their comparisons. The top two rows represent WBCs subtypes from the PBC dataset along with their corresponding bounding boxes and detection percentage values. The middle two rows show the WBCs subtypes from the LISC dataset [70], also accompanied by bounding boxes and detection confidence values in percentage information.

Similarly, the last two rows of Fig. 11 show the WBCs subclasses from the Raabin-WBC dataset, along with bounding boxes and detection accuracy percentage values. A comparison was made with 15 existing algorithms, as shown in Table 13, across three distinct datasets. The experimental results offer valuable insights into the effectiveness and efficiency of the proposed method when compared to SOTA approaches. As an example, Zhang et al. [76] utilized an average ensemble model to improve classification performance, achieving a testing accuracy of 90.1% on the PBC Dataset. Wang et al. [77] employed both an SSD and an enhanced version of YOLO for the recognition of WBCs in microscopic blood smear images. Kimura et al. [78] presented a blood cell classification system that utilizes extreme gradient boosting (XGBoost) to help medical experts provide information for accurate decision-making. Chen et al. [40] used ResNet and DenseNet models with a Spatial and Channel Attention Module (SCAM) to achieve accurate classification of WBCs. Together, the visual and quantitative analyses offer a comprehensive evaluation of our classification framework. Fig. 11 displays our classification results, and Table 14 compares the metrics with those of SOTA methods. In Table 14, the evaluation metrics are recorded in terms of recall, specificity, precision, and accuracy using the PBC benchmark dataset. We empirically determined that the proposed method achieved the highest accuracy of 99.83% on this dataset. The proposed model surpasses the other methods across all indicators in terms of specificity, recall, precision, and accuracy. The proposed model achieved substantial improvements of 9% in specificity and 3% in accuracy. In summary, our model clearly outperforms the existing SOTA approaches. The proposed technique exhibited a significant increase in accuracy for this dataset after integrating the DA module. This enhancement highlighted the effectiveness of the DA mechanism in further enhancing the performance and accuracy of the model.

Table 13
Comparison of the proposed Efficient DA-Coupled CNN with existing methods, in terms of overall accuracies, on PBC, LISC and Raabin-WBC benchmark datasets.

Method/Reference	Year	Dataset	Techniques used	Overall Accuracy (%)
Wang, Qiwei et al. [77]	2019	PBC	SSD and an improved YOLOv3	90.09
Kimura, Konobu et al. [78]	2019		CNN with XGBoost algorithm	96.20
Zhang, Ruitao et al. [76]	2022		RCM-Net	99.63
Liu, Rui et al. [75]	2022		Employed ResNeXt-50-32 × d	96.83
Our Proposed	–		**Efficient DA-Coupled CNN**	**99.83**
Kutlu et al. [4]	2020	LISC	R–CNN	97.52
Chen, Hua et al. [40]	2022		ResNet and DenseNet with SCAM	97.96
Jung, Changhun et al. [79]	2022		W-Net with GCGAN	97.00
B, Kishore et al. [37]	2023		CNN Optimized by Particle Swarm Optimization	99.2
Bairaboina et al. [74]	2023		W-Net	99.16
Balasubramanian et al. [80]	2023		U-Net-SVM	98.62
Our proposed	–		**Efficient DA-Coupled CNN**	**99.35**
Rivas-Posada, E. et al. [81]	2022	Raabin-WBC	Meta Learning (Mel) Model	96.40
Rao, Bairaboina et al. [82]	2023		MobilenetV3-ShufflenetV2	99.00
Balasubramanian et al. [80]	2023		U-Net-SVM	99.45
Leng, Bing et al. [25]	2023		Detection Transformer (DETR)	96.10
Katar, Oguzhan et al. [36]	2023		Vision Transformer (ViT)	99.40
Our proposed	–		**Efficient DA-Coupled CNN**	**99.60**

Additionally, to offer a more intuitive representation of the performance of our proposed model, we present its detailed confusion matrix in Fig. 10. This visualization highlights the comprehensive classification capability of the model, as demonstrated by a misclassification rate of only 0.02 %.

5. Discussion and limitations

In this work, we came up with a methodology that combines CNN with DA to address some problems of WBCs detection and classification area and provide potential directions for future research. The decision to integrate DA into the CNN was informed by two factors; overcoming computational costs and model sizes. Inclusion of the proposed method improved efficiency in dealing with microscopic images containing

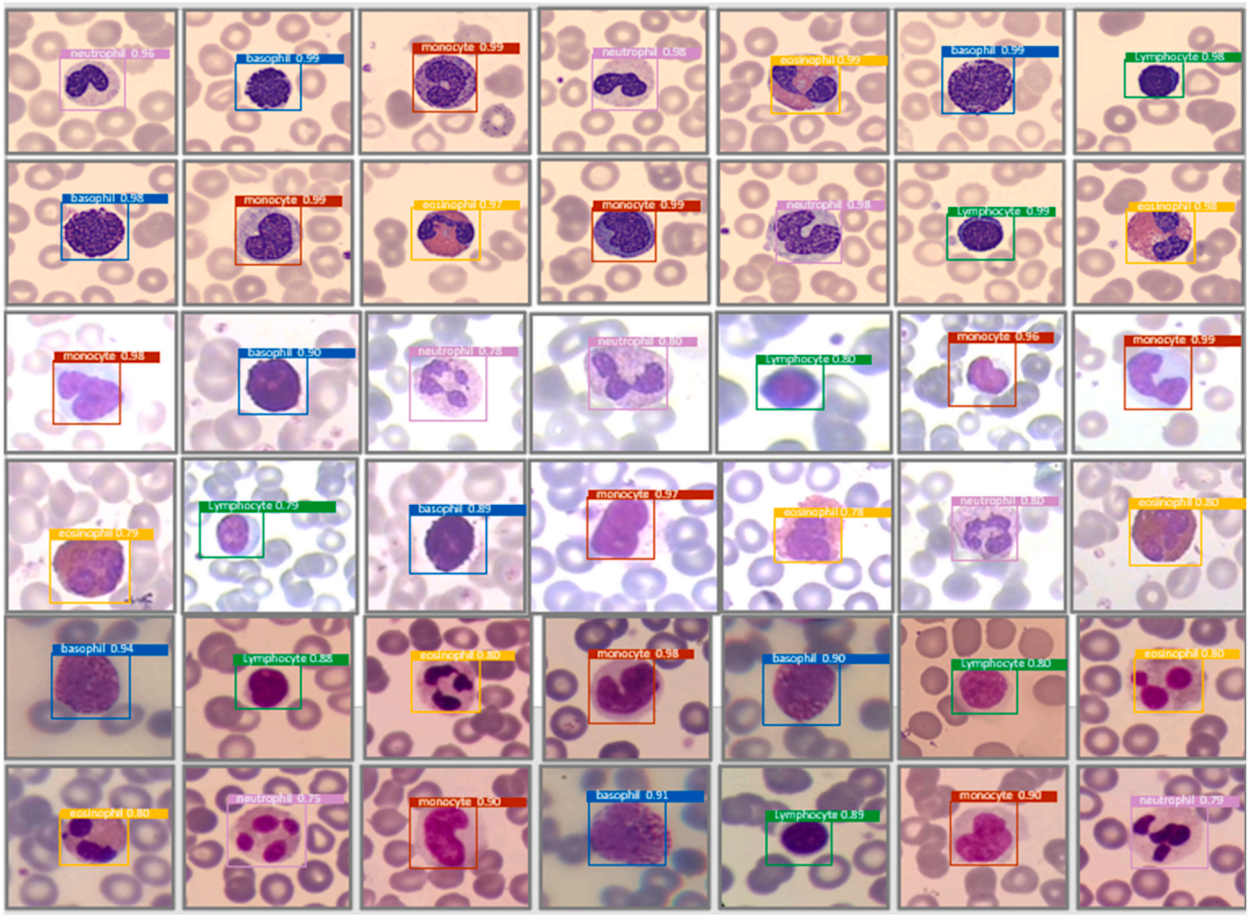


Fig. 11. Visual results of the proposed technique on test images obtained from PBC, LISC and Raabin-WBC test sets.

Table 14
Comparative analysis of the proposed method with SOTA methods in terms of Specificity, Recall, Precision, and Accuracy.

Methods	Specificity (%)	Recall (%)	Precision (%)	Accuracy (%)
[77] 2019	0.89	0.92	0.89	90.09
[78] 2019	0.90	0.95	0.90	96.20
[76] 2022	0.88	0.93	0.89	99.63
[75] 2022	0.88	0.95	0.92	96.83
[25] 2023	**0.97**	0.96	0.96	96.10
[74] 2023	**0.97**	**0.98**	0.98	98.87
Proposed	**0.97**	**0.98**	**0.99**	**99.83**

WBCs through incorporation of DA. This advancement is important in ensuring that real-time health care systems have timely and accurate diagnoses. Moreover, reduction in model size makes it useful for devices which are resource constrained hence allowing accessibility even when there are limited computational resources available at hand. Another advantage worth mentioning about our method is its capability to improve detection accuracy rates; this is achieved through making the model more sensitive towards features or patterns specific for WBCs as they were earlier compared against each other during training phase only using their common characteristics such as shape color etc., without any knowledge about biological structure behind them. According to experimental results presented in Table 13, our model achieved higher overall accuracy rates: 99.83% PBC dataset; 99.35% LISC dataset; 99.60% Raabin Dataset. The promising outcomes demonstrated here suggest that the novel methodology described may surpass current techniques, emphasizing its potential value. The proposed framework shows early signs of significant ability to move the field forward in an important way. Further research is warranted to explore this approach and understand its capabilities more fully.

In the proposed framework, elimination of additional preprocessing steps and cell pre-segmentation is a crucial aspect of our methodology. Preprocessing has traditionally been a very involved process, which can take a long time and can introduce mistakes. We have just reduced all these processes creating a less difficult diagnostic process with fewer human errors possible. This is especially important in the field of medical diagnostics where accuracy and speed are everything. Clinical settings require real-time processing for fast diagnosis and quick decision making. One another restriction is the computational demand of Faster R-CNNs, especially with high-resolution medical images. In the proposed method, the key limitation is using Faster R–CNN for WBC detection. Although Faster R–CNN is an accurate and robust object identification model, it may not be the best choice for real-time applications because it requires too much computation power in resource-limited contexts. Within an input image, a Faster R–CNN proposes regions using an RPN. Then these proposals pass on to be classified and bounding-box regressed by deep CNNs.

Although this method gives consistent results, it is computationally intensive and may not be suitable for real time processing. This limitation can reduce the ability of the model to produce timely results which is important in emergency medicine or when dealing with large patient datasets. For further studies, more computationally efficient object detection models should be considered or the current Faster R–CNN model should be modified to run in real time so as to address this problem. Furthermore, utilizing hardware acceleration or cloud-based processing may aid in mitigating the model computing demands and improving its real-time performance. Recognizing this restriction, we acknowledged the necessity of balancing accuracy and efficiency in medical diagnostic systems, and we try to enhance our method for real-time application in resource-constrained healthcare settings.

6. Conclusion and future work

The accurate detection of WBCs in microscopic images of blood is a technically challenging task that required advanced algorithms, ML techniques, and robust analytical frameworks. Therefore, effective methods are needed to simplify the process of recognizing WBCs. With this goal in mind, we proposed a DA-based CNN model that can detect overlapping and complexly structured WBCs/leukocytes in blood microscopic images. This study successfully proposed a robust DA-based CNN framework to detect and categorize WBCs in microscopic blood smear images. The proposed method exhibited considerable improvements in the accuracy of 99.83%, 99.60%, and 99.35 on the PBC [69], LISC [70] and Raabin-WBC [71] datasets, respectively. The utilization of the DA mechanism greatly enhanced the model's ability to focus on relevant spatial and channel information, enabling more precise classification outcomes. Furthermore, the incorporation of DCGAN data augmentation successfully addressed the challenges posed by the insufficiency and imbalance of WBC samples.

In the future, we plan to optimize advanced detection algorithms to improve their performance and real-time deployment in the field of medicine and assist hematologists and doctors. Moreover, the use of high-quality and large amounts of medical data is likely to improve the performance of the proposed technique. Therefore, we will also investigate advanced detection algorithms and compare their performance on smart and portable devices.

CRediT authorship contribution statement

Siraj Khan: Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Muhammad Sajjad:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Naveed Abbas:** Writing – review & editing, Validation, Methodology, Formal analysis. **José Escorcia-Gutierrez:** Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Formal analysis. **Margarita Gamarra:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Khan Muhammad:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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